

6 Further Matrix Algebra

What students need to learn:

Linear transformations of column vectors in two and three dimensions and their matrix representation.

Combination of transformations. Products of matrices.

Transpose of a matrix.

Evaluation of 3×3 determinants.

Inverse of 3×3 matrices.

The inverse (when it exists) of a given transformation or combination of transformations.

Eigenvalues and eigenvectors of 2×2 and 3×3 matrices.

Reduction of symmetric matrices to diagonal form.

Extension of work from FP1 to 3 dimensions.

The transformation represented by $\mathbf{AB}$ is the transformation represented by $\mathbf{B}$ followed by the transformation represented by $\mathbf{A}$.

Use of the relation $(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$.

Singular and non-singular matrices.

Use of the relation $(\mathbf{AB})^{-1} = \mathbf{B}^{-1} \mathbf{A}^{-1}$.

Normalised vectors may be required.

Students should be able to find an orthogonal matrix $\mathbf{P}$ such that $\mathbf{P}^T \mathbf{A} \mathbf{P}$ is diagonal.

Unit M1 Mechanics 1

GCE AS and GCE Mathematics, GCE AS and GCE Further Mathematics and GCE AS and GCE Further Mathematics (Additional) AS optional unit

M1.1 Unit description

Mathematical models in mechanics; vectors in mechanics; kinematics of a particle moving in a straight line; dynamics of a particle moving in a straight line or plane; statics of a particle; moments.

M1.2 Assessment information

Prerequisites

Students are expected to have a knowledge of C1, its preambles and associated formulae and of vectors in two dimensions.

Examination

The examination will consist of one 1½ hour paper. It will contain about seven questions with varying mark allocations per question which will be stated on the paper. All questions may be attempted.

Calculators

Students are expected to have available a calculator with at least the following keys: $+$, $-$, $\times$, $\div$, π , x^2 , $\sqrt{x}$, $\frac{1}{x}$, x^y , $\ln x$, e^x , sine, cosine and tangent and their inverses in degrees and decimals of a degree, and in radians; memory. Calculators with a facility for symbolic algebra, differentiation and/or integration are not permitted.

Formulae

Students are expected to know any other formulae which might be required by the specification and which are not included in the booklet, *Mathematical Formulae including Statistical Formulae and Tables*, which will be provided for use with the paper. Questions will be set in SI units and other units in common usage.

Students will be expected to know and be able to recall and use the following formulae:

$$\text{Momentum} = mv$$

$$\text{Impulse} = mv - mu$$

For constant acceleration:

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

$$s = vt - \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

$$s = \frac{1}{2}(u + v)t$$

1 Mathematical Models in Mechanics

What students need to learn:

The basic ideas of mathematical modelling as applied in Mechanics.

Students should be familiar with the terms: particle, lamina, rigid body, rod (light, uniform, non-uniform), inextensible string, smooth and rough surface, light smooth pulley, bead, wire, peg. Students should be familiar with the assumptions made in using these models.

2 Vectors in Mechanics

What students need to learn:

Magnitude and direction of a vector. Resultant of vectors may also be required.

Students may be required to resolve a vector into two components or use a vector diagram. Questions may be set involving the unit vectors $\mathbf{i}$ and $\mathbf{j}$.

Application of vectors to displacements, velocities, accelerations and forces in a plane.

Use of $\text{velocity} = \frac{\text{change of displacement}}{\text{time}}$

in the case of constant velocity, and of $\text{acceleration} = \frac{\text{change of velocity}}{\text{time}}$

in the case of constant acceleration, will be required.

3 Kinematics of a particle moving in a straight line

What students need to learn:

Motion in a straight line with constant acceleration.

Graphical solutions may be required, including displacement-time, velocity-time, speed-time and acceleration-time graphs. Knowledge and use of formulae for constant acceleration will be required.